Harmonic Analysis Dashboard — User Guide

Review harmonic-voltage results, compare cases, and prepare chart exports.

Working draft — pending MPE review

Open the Harmonic Analysis Dashboard

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1. Introduction

The Harmonic Analysis Dashboard is a browser-based tool for reviewing harmonic-voltage distortion results supplied as Excel workbooks. Use it to compare result datasets and locations, inspect harmonic spectra, display selected harmonic limits, review calculated voltage total harmonic distortion (THD), define reusable case selections, and download charts or numerical data.

Open the application at the link on the cover. Use a modern browser. The hosted workflow does not require local Python, a batch file, or access to a development folder. If the hosting service displays an access restriction or asks the app to start, follow the on-screen instructions. The dashboard does not add a separate password system.

The dashboard presents and calculates values from supplied files. It does not run PSCAD simulations, verify modelling assumptions, select an applicable limit automatically, or issue an engineering compliance decision. You remain responsible for selecting the correct dataset, location, limit source and assessment basis. A value above a selected curve at one harmonic order is a local numerical exceedance only. It is not, by itself, a complete compliance conclusion.

For a first analysis, follow Section 2. Use Sections 3–14 to learn the interface, workbook rules, chart behaviour, saved case selections, exports and recovery steps.

2. Quick Start

Use this procedure to reach the first comparison chart.

1. Open the hosted dashboard from the cover link.
2. In the left **Analysis setup** sidebar, expand **Add Excel workbooks**. Upload at least one result workbook. You can upload separate **.xlsx** files, a **.zip** bundle containing workbooks, or both. Add a case-mapping workbook and limit workbook when you need them; both are optional.
3. Select a **Result dataset** and a **Busbar / location**. The location names come from worksheet names in that result workbook.
4. Under **Limit series**, choose the relevant series or **No limit**. If you choose a series, choose **Bars** or **Dashes** under **Limit display**.
5. In **Groups**, choose a standard group or **All**. Use **Cases to compare** to reduce the case list. The spectrum and separate THD chart update for the selected cases.
6. Hover over a marker to inspect its value and case description. Use **Download plotted data (CSV)** for the current numerical spectrum, the chart camera icon for one PNG, or **Download charts** for a configurable ZIP.

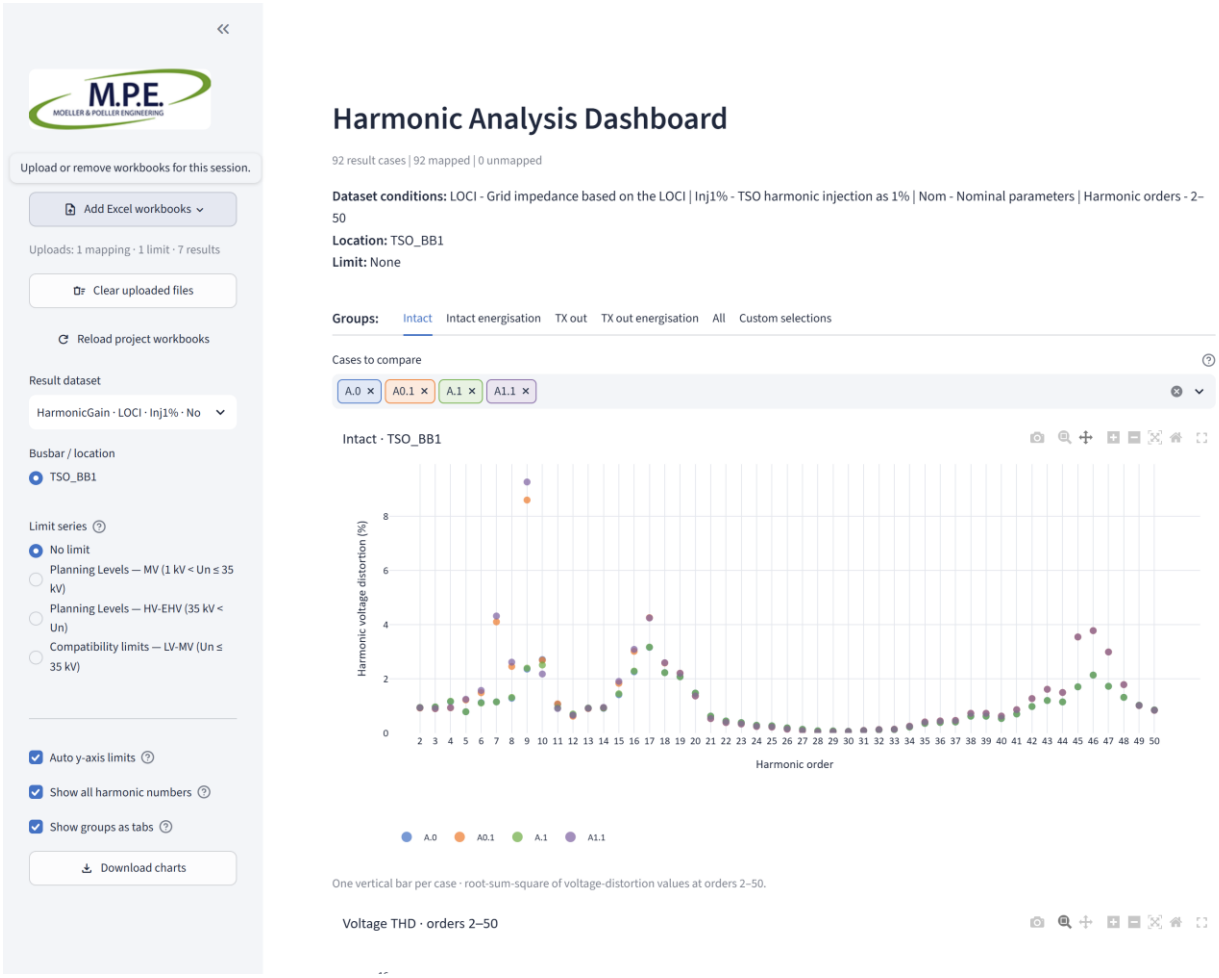
See Section 4 for workbook formats, Section 10 for description-based case selections, and Section 12 for the ZIP workflow.

3. Application Interface

The workspace has three main areas.

Area	What it contains
Analysis setup — left sidebar	Workbook upload and reset controls; dataset and location selection; limit source, series and display; live axis settings; harmonic labels and group layout; Download charts.
Context and status — above the charts	Result and mapping counts; dataset conditions decoded from the mapping workbook where available; harmonic-order range; current location and limit; Data validation and sources.
Analysis views — main workspace	Group views, All, conditional Unmapped, and Custom selections; Cases to compare; harmonic spectrum; THD chart; selected-case descriptions; plotted-data CSV action.

Figure 1 — Dashboard overview using the example workbook set.



The available controls depend on the loaded files. A valid result workbook is required. Mapping and limit workbooks are optional. Without mapping, the dashboard still provides **All** and **Custom selections**, but a text rule cannot find cases without mapped descriptions. See Sections 4–6.

The **Data validation and sources** expander lists the active source files and reports validation by location. Open it when a chart is missing or a warning appears. Section 13 explains how to act on its messages.

4. Input Data and Workbook Requirements

4.1 Workbook roles and filenames

The dashboard identifies workbook roles from each workbook's filename. This rule also applies to .xlsx files inside a ZIP archive.

Role	Recognised filename	Required?	Purpose
Result	Any .xlsx file not recognised as mapping or limits. Total distortion_...xlsx is common, but not required.	Yes — at least one	Harmonic values for cases and locations.
Case mapping	Cases_description...xlsx; the prefix is case-insensitive.	Optional — one active mapping	Case groups, descriptions and condition-token definitions.
Limits	Limits_...xlsx; the prefix is case-insensitive.	Optional — one or more sources	Named harmonic-limit series. Choose No limit when no limit is needed.

Use distinct workbook filenames. The dashboard rejects duplicate basenames, including duplicates stored in different ZIP folders. It does not require a particular text after a recognised mapping or limits prefix.

4.2 Result workbook

Each worksheet represents one busbar or measurement location. The worksheet name appears in **Busbar / location**. The first row contains headers. The first column must be **Harmonic Order**. Each other column is a case code. The case code must be unique in that worksheet and must be numeric data or blank below the header.

Illustrative structure — values are not study results.

Harmonic Order	A.0	A0.1
2	1.20	0.80
3	0.45	0.52
5	0.13	0.09

Enter harmonic-voltage distortion values as percentages. For example, enter 1.20 for 1.20%. Do not multiply the values by 100. Harmonic orders must be numeric and unique. Do not leave a case header empty. A value may be blank, but blank values can prevent a THD result. The spectrum uses the harmonic orders in the sheet. It is not limited to order 50.

4.3 Case-mapping workbook

The mapping workbook must contain two worksheets:

- **Cases** — exact columns: **Case number**, **Group**, **Description**.
- **Conditions** — exact columns: **Token**, **Description**.

Case numbers must match result-workbook headers exactly. Group labels determine the standard group views. The standard groups are **Intact**, **Intact energisation**, **TX out**, and **TX out energisation**.

Illustrative structure — values are not study results.

Case number	Group	Description
A.0	Intact	with 154 kV Filters and WTGs
A0.1	Intact	no 154 kV Filters and with WTGs

Token	Description
LOCI	Grid impedance based on the LOCI
Inj1%	TSO harmonic injection as 1%

Other group labels produce a warning. The dashboard does not infer a standard group from a case code. Unmapped result codes remain available under **Unmapped** and **All**; they are not discarded. Blank descriptions produce a warning and cannot match a custom text rule. The **Conditions** sheet can explain tokens found in dataset filenames. Unknown tokens remain unexplained.

4.4 Limits workbook

The first worksheet must contain a **Harmonic Order** column and at least one named limit series. The parser searches the first ten rows for the harmonic-order heading and supports multi-row or merged series headings.

Illustrative structure — values are not study results.

Harmonic Order	Example planning series	Example compatibility series
2	2.0	2.0
3	2.0	2.0
5	1.5	1.5

This table only shows the relationship between order and series. It does not identify which engineering curve applies to a location. Choose the limit source and series deliberately. The dashboard does not assign a voltage class automatically.

The dashboard aligns a limit with the same numeric harmonic order. It does not align by worksheet row. A blank or missing limit at an order is undefined. The dashboard does not replace it with zero, interpolate it, or extrapolate it.

4.5 Upload constraints

The upload control accepts **.xlsx** workbooks and **.zip** archives. It does not accept .xls or .csv as input. ZIP archives can contain workbooks in subfolders. The dashboard reads supported workbooks from the archive without extracting them to a user folder.

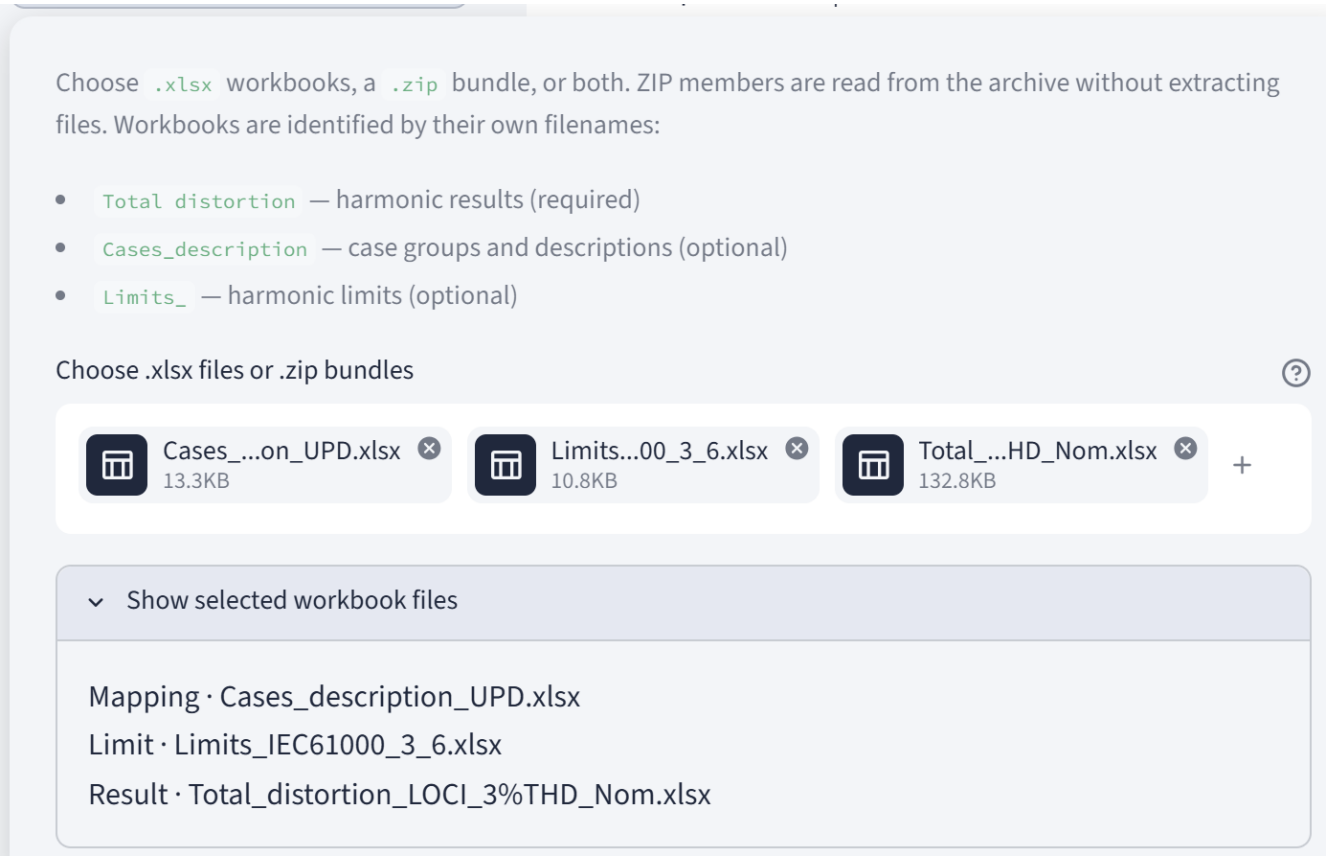
One ZIP archive can contain up to 100 entries and up to 100 MiB of total uncompressed content. This archive limit is separate from the per-file upload limit displayed by the browser control. Unsupported ZIP entries and Excel lock files are ignored. Encrypted or corrupt archives are rejected.

5. Loading and Managing Workbooks

5.1 Add workbooks

1. Expand **Add Excel workbooks** in the sidebar.
2. Choose one or more .xlsx workbooks, one or more .zip bundles, or a mixture.
3. Check the selected-file list when you need to confirm how each workbook was classified.
4. Select the result dataset to analyse. Add mapping and limits only when the analysis needs them.

Figure 2 — Upload roles and the selected workbook list.



Uploads stay in the current Streamlit session. The dashboard does not save an uploaded workbook as a permanent project input file. You do not need a shared folder.

5.2 How uploads interact with project workbooks

An uploaded mapping workbook replaces any preloaded mapping workbook for the current session. Uploaded result and limit workbooks are added to the available choices. If an uploaded file has the same filename as a preloaded file, the uploaded file takes precedence. Only one mapping workbook can be active.

When uploads are cleared, any remaining preloaded workbooks become available again.

5.3 Clear or reload

Control	Action
Clear uploaded files	Removes workbooks uploaded in the current session. It does not delete project input files.
Reload project workbooks	Re-scans the app's preloaded inputs and clears the related data cache. It keeps current session uploads.

For a hosted user, Reload project workbooks does not provide access to a server folder. Upload changes can reset dataset or case selections when the earlier state no longer applies.

5.4 Session boundary

Uploaded files and saved custom selections are session data. A new session can require you to upload workbooks again and import any rules you need. Section 10.5 explains how to back up saved selections.

6. Configuring the Analysis

6.1 Choose a result dataset

Use Result dataset to choose a result workbook. The dashboard uses filename tokens to create a shorter label. For example, a workbook beginning with Total distortion_ does not repeat that prefix in the selector. When matching definitions are present in the mapping workbook's Conditions sheet, the context line explains those tokens. Unknown tokens are not translated automatically.

The selector may appear as a list or a dropdown, depending on the number of available choices. The label and selection meaning stay the same.

6.2 Choose a location

Use Busbar / location to choose one worksheet in the selected dataset. The choices and number of harmonic rows can differ by dataset. The dashboard does not combine data from different locations in the live chart.

6.3 Choose a limit

When more than one limits workbook is available, first select a Limit source. Then select a Limit series, or choose No limit. A source can contain several planning or compatibility series. The dashboard does not decide which one applies.

The live dashboard remembers the selected limit series separately for each location and limit source. A location that has no saved choice starts with the currently active series. Changing the choice changes the displayed reference, not the result values.

When a limit is active, use Limit display to choose:

- Bars — light translucent vertical bars show the limit range at each harmonic order.
- Dashes — short red horizontal markers show the limit value at each harmonic order.

6.4 Set the y-axis and harmonic labels

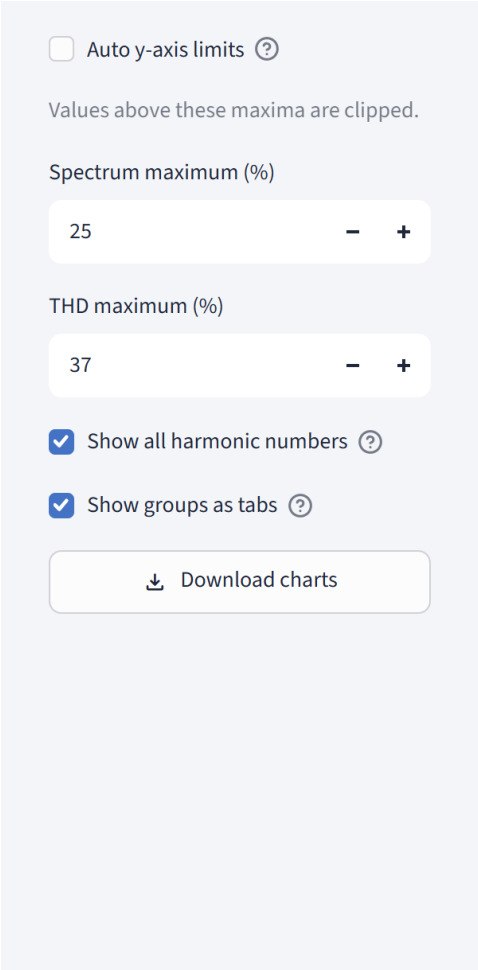
Auto y-axis limits is on by default. In this mode, each live chart fits its plotted values. Turn it off to enter separate Spectrum maximum (%) and THD maximum (%) values.

Live manual maxima are saved per result dataset. They apply across that dataset's locations and groups. Spectrum and THD use separate maxima. Both axes start at zero. If a fixed maximum is below a plotted value, the chart clips the

value and displays a warning. It does not change the source value or calculation. The initial manual maxima are rounded up from the dataset values and the selected limit values where relevant.

Show all harmonic numbers is on by default. When it is off, the axis uses fewer labels; the data points remain. **Show groups as tabs** is on by default. Turn it off to show the group sections vertically.

Figure 3 — Manual axis maxima and the default-on display controls.



7. Selecting and Comparing Cases

7.1 Group views

With a mapping workbook, the dashboard provides the four standard groups, **All**, **Custom selections**, and **Unmapped** when result codes are missing from the mapping. Without mapping, **All** and **Custom selections** remain. The text-based custom rule cannot match case descriptions until mapping data is available.

All combines cases available at the selected location. It is a live comparison view. Group assignment comes from the mapping workbook, not from case-code order or numbering.

7.2 Use the case selector

Cases to compare selects all eligible cases by default. Remove a case chip to hide that case. Open the selector to choose individual codes or use **Select all** and **Clear all**. If no cases remain selected, the view shows an instruction instead of a chart.

Case colours remain stable when you reduce the selection. Case choices are remembered for the current dataset, location and group during the session. Switching to another context does not apply the previous context's case-chip state.

7.3 Select a case from the chart

Click a spectrum data point to narrow the case selection to that case. To restore all available cases in the group, **Shift-click** the selected point. In **Custom selections**, this restores all current rule matches.

Use the case chips when you want to make several changes. A Plotly legend action changes trace visibility; it does not replace the saved case selection.

7.4 Read case descriptions

Mapped case codes and descriptions appear in the spectrum hover. The **Selected case descriptions** table lists the current selected cases with **Case** first and **Description** second. An unmapped result is identified by its exact case code. The dashboard does not invent a description or group.

8. Harmonic Spectrum Analysis

The spectrum shows harmonic voltage distortion **at each harmonic order for each selected case**. It is not a THD chart.

- Horizontal axis — **Harmonic order**.
- Vertical axis — **Harmonic voltage distortion (%)**.
- Each case — coloured markers. The markers are not joined by lines.
- Order range — the orders in the selected result worksheet, including orders above 50 when supplied.

Figure 4 — Spectrum markers, selected limit dashes and a harmonic-order hover.



8.1 Read the hover

Hover over a case marker. The hover shows the harmonic order and the hovered case's code, value and mapped description where available. When a limit is selected, the hover can also show other selected cases above that limit at the same order. Those values are ordered from highest to lowest and highlighted in red. The hovered case is visually distinguished from the other affected cases. The selected limit appears last and uses a red dash symbol.

A blank limit at an order is undefined, not zero. Dense views may summarise other exceedances to keep the hover readable.

8.2 Interpret the limit display

Bars and dashes are two ways to display the same selected limit series. They do not change the values. A case value above the selected limit at one harmonic is a local curve exceedance at that order. It does not show that all other orders exceed or pass, and it does not confirm that the chosen limit applies to the study.

8.3 Use the Plotly chart controls

The chart toolbar provides the visible Plotly actions, which can include zoom, pan, autoscale or reset, and **Download plot as a PNG** on the camera icon. Hover over an icon to read its tooltip. Use the legend to show or hide traces.

The camera action saves the chart currently displayed. It is separate from **Download plotted data (CSV)** and the multi-chart ZIP in Section 12.

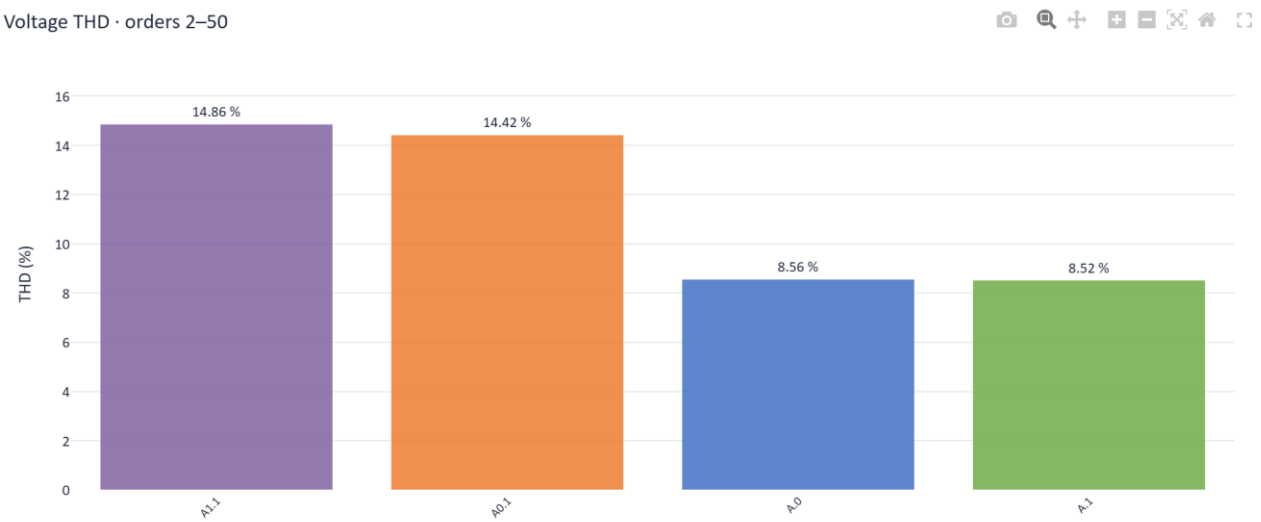
8.4 Check axes and missing values

If a manual y-axis maximum is too small, the chart clips higher values and shows a warning. Increase the maximum or restore **Auto y-axis limits**. A blank source value remains blank; it is not plotted as zero.

9. Voltage THD Analysis

The separate **Voltage THD** chart appears below the harmonic spectrum for a selected group. It uses one vertical bar per case. The bars are sorted from highest calculated THD to lowest. Each case keeps its case colour. The bar labels show two decimal places. Hover can show greater precision and the case description.

Figure 5 — Separate voltage THD chart with values to two decimal places.



For each case, the dashboard calculates:

THD_u (%) = √(Σ u_h(%)²), for h = 2 to H, where H = the smaller of 50 and the highest integer harmonic order available.

The calculation uses the supplied harmonic-voltage percentages from orders 2 through H. The spectrum can display orders above 50; this THD calculation stops at order 50.

Every order and numeric case value in the required calculation range must be present. The dashboard does not replace a missing order or value with zero and does not calculate a partial THD. It omits that case's bar and lists missing orders. The spectrum can still be available. If no selected case has complete data, the dashboard shows a message instead of bars.

The selected harmonic limit does not apply to THD. The THD chart does not imply a THD limit or compliance result.

10. Custom Case Selections

10.1 What a custom selection does

A custom selection is a saved description-based rule. It is not a fixed list of case codes. The rule searches the mapping workbook's **Description** column and matches only result case codes available in the current dataset and location. The mapping workbook is required for text matches.

The same rule is recalculated when you change the dataset or location. It selects cases across standard groups. Use the case chips to narrow the matched cases further.

10.2 Build a rule

1. Open **Custom selections** and choose **New selection**.
2. Enter a **Name** or leave it blank. If blank, the dashboard generates a name from the rule.
3. Add one or more phrases to the relevant boxes. Type a phrase and press Enter to add it as a chip.
4. Review the resulting rule and matched-case count before you select **Save selection**.

Box	Rule
Must contain ALL	Every phrase in this box must appear in the case description.
Must contain ANY	At least one phrase must appear when this box contains entries. Leave it empty to skip this condition.
Must NOT contain	Exclude a case if any phrase in this box appears in its description.

All non-empty boxes are combined with AND. Phrases within the **ANY** and **NOT** boxes act as alternatives. A rule must contain at least one phrase.

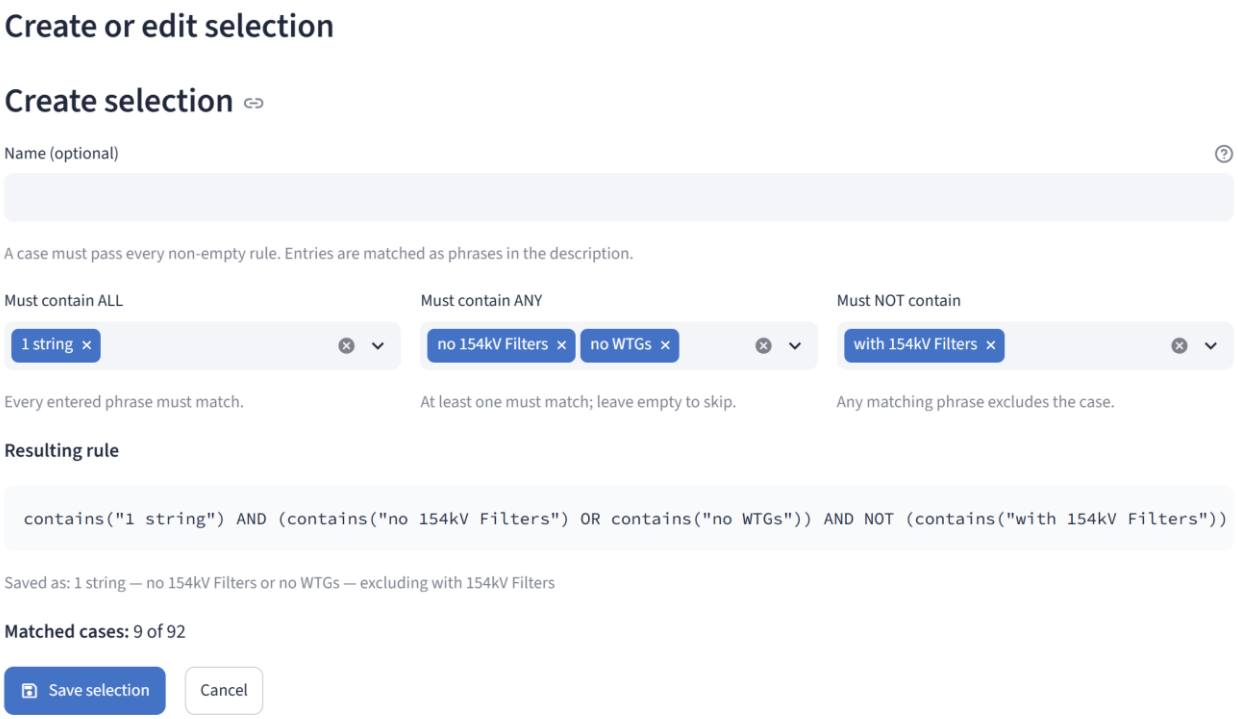
Matching is case-insensitive and uses complete ordered phrases. The dashboard normalises supported variants such as **154 kV** and **154kV**, **Filter** and **Filters**, and **string** and **strings**. It does not match **5 strings** to **15 strings** by substring. It does not infer synonyms. Rules search descriptions only; they do not search case codes, group names or workbook filenames. Enter separate phrases as separate chips. A comma in a description does not split a phrase automatically.

Illustrative logic — this is not a result count.

Box	Phrase
ALL	5 strings
ANY	no 154kV Filter; no WTGs
NOT	BB2

This means: the description contains **5 strings**, and it contains either **no 154kV Filter** or **no WTGs**, and it does not contain **BB2**. A real match count depends on the current mapping, dataset and location.

Figure 6 — Create a rule, read the expression and review matched cases.



10.3 Preview and save

The editor displays a readable **Resulting rule** and **Matched cases** count for the current dataset and location. Cases with no description cannot match. Select **Save selection** to add the rule to the current session. Select **Cancel** to discard an unsaved edit. The workflow does not use a separate Apply button.

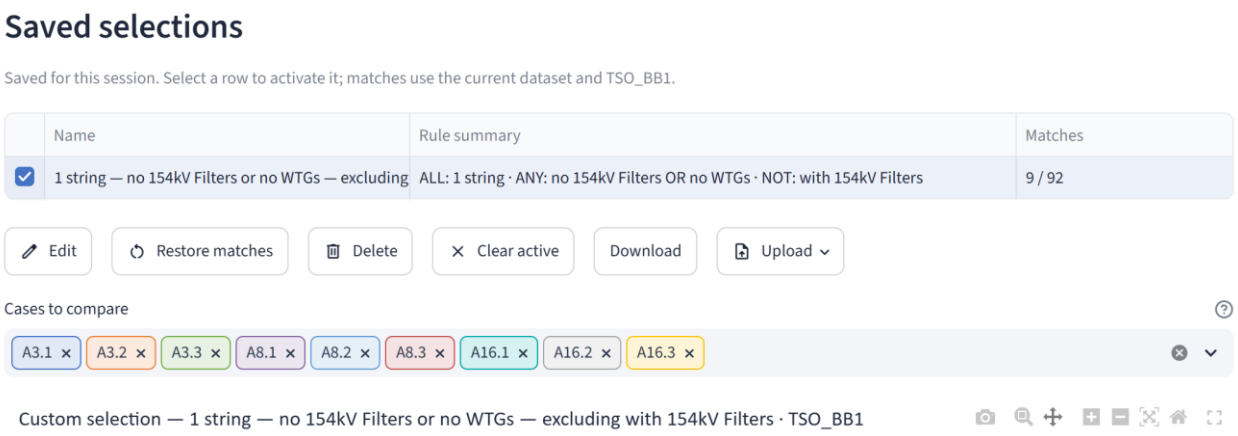
10.4 Activate and manage saved selections

The **Saved selections** table shows each rule's name, rule summary and current match count. Select a row to activate it. The dashboard then updates the case chips and charts. Only one saved selection can be active at a time.

Action	Result
Edit	Opens the selected rule for changes. Changes take effect after Save selection.
Restore matches	Resets manual case-chip reductions to all cases that match the active rule.
Clear active	Clears the active rule. It does not delete the saved rule.
Delete	Removes the saved rule from this session after confirmation. It does not change a workbook.

If a rule has no matches, the dashboard shows its no-match message. It does not create a zero-valued chart.

Figure 7 — Saved rule row, management actions and matched case chips.



10.5 Back up or import rules

Saved selections belong to the current session. Select **Download** to save a JSON backup named **murakami_saved_selections.json**. The file contains the rule library, not uploaded workbooks.

To reuse rules, select **Upload**, choose a valid JSON backup, then select **Import selections**. Import replaces the rule library in the current session; it does not merge lists. An invalid JSON file leaves the current library unchanged. Importing rules does not guarantee matches in another dataset. The case descriptions must contain the rule phrases.

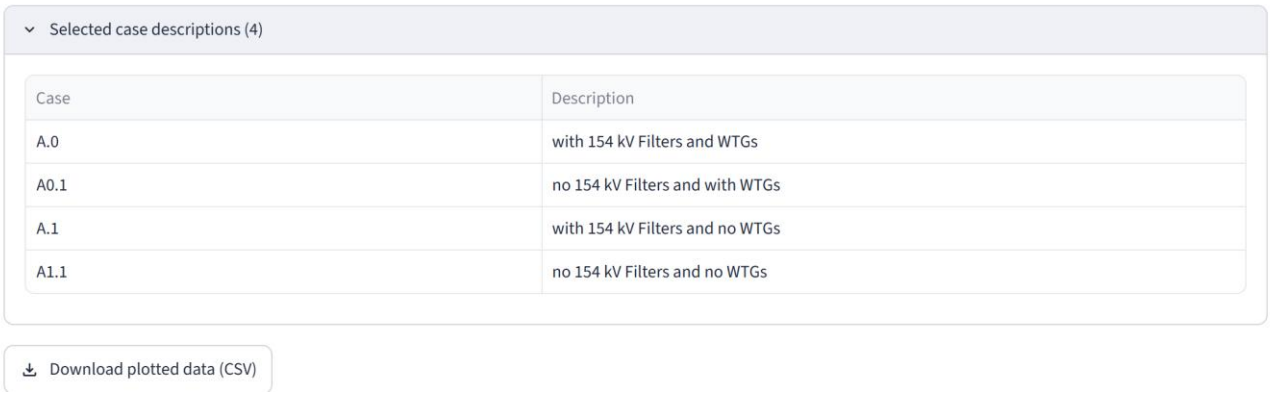
11. Downloading Individual Results

Use these actions for one current view or its displayed numerical spectrum.

Action	Output	Scope
Chart toolbar camera	PNG image	The currently displayed spectrum or THD chart.
Download plotted data (CSV)	CSV table of selected harmonic orders and case values; includes the selected limit column when applicable.	Current dataset, location, group and case selection.
Download charts	ZIP archive of selected chart sets, with optional THD images and case-description CSV files.	Batch plan configured in Section 12.

The plotted-data CSV preserves the numeric spectrum values. It does not add a calculated THD column. The chart ZIP does not contain the plotted harmonic-data CSV.

Figure 8 — Case descriptions and the plotted-data CSV action.



12. Batch Chart Export

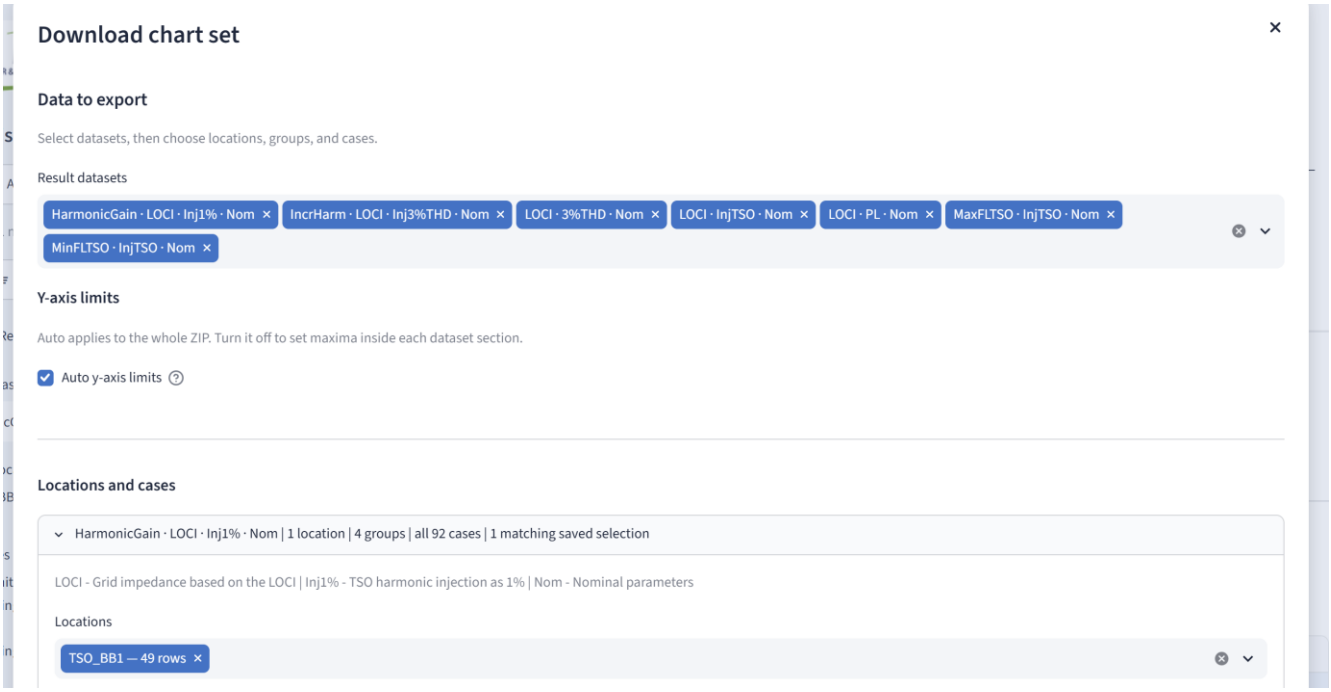
Select **Download charts** in the sidebar to open **Download chart set**. Work through its sections in order: **Data to export**, **Chart settings**, **ZIP layout**, and **Review and download**.

Export options are separate from the live chart state in some areas. In particular, the export axes, harmonic labels and selected chart sets are controlled in the dialog.

12.1 Select result datasets

The **Result datasets** selector starts with all available datasets selected. Use **Select all** or **Clear all** in multiselects when needed. Expand a dataset row to set its locations, groups and cases. A dataset only produces charts for its selected locations.

Figure 9 — Dataset selection and the nested locations, groups and case options.



12.2 Select locations, groups and cases

In each dataset, select one or more **Locations** and **Case groups**. A case must be present at every selected location to be eligible. When you select several locations, only their shared eligible cases can be selected.

With mapping, choose from the standard groups and **Unmapped** when required. **All** is a live comparison view; it is not another standard export group. Without mapping, the export uses a single **Cases** group based on the result case codes.

All cases is selected by default. Clear it to select individual case codes for the chosen groups. The chart-set preview changes when you change the locations, groups or case choices.

12.3 Add saved custom selections

When a mapping and saved rules are available, the dataset section provides **Additional custom chart sets** and **Saved selections to include**. Available rules start selected. Each rule that matches eligible cases adds a separate chart set for each selected location, alongside the standard group charts. A rule with no eligible cases is skipped.

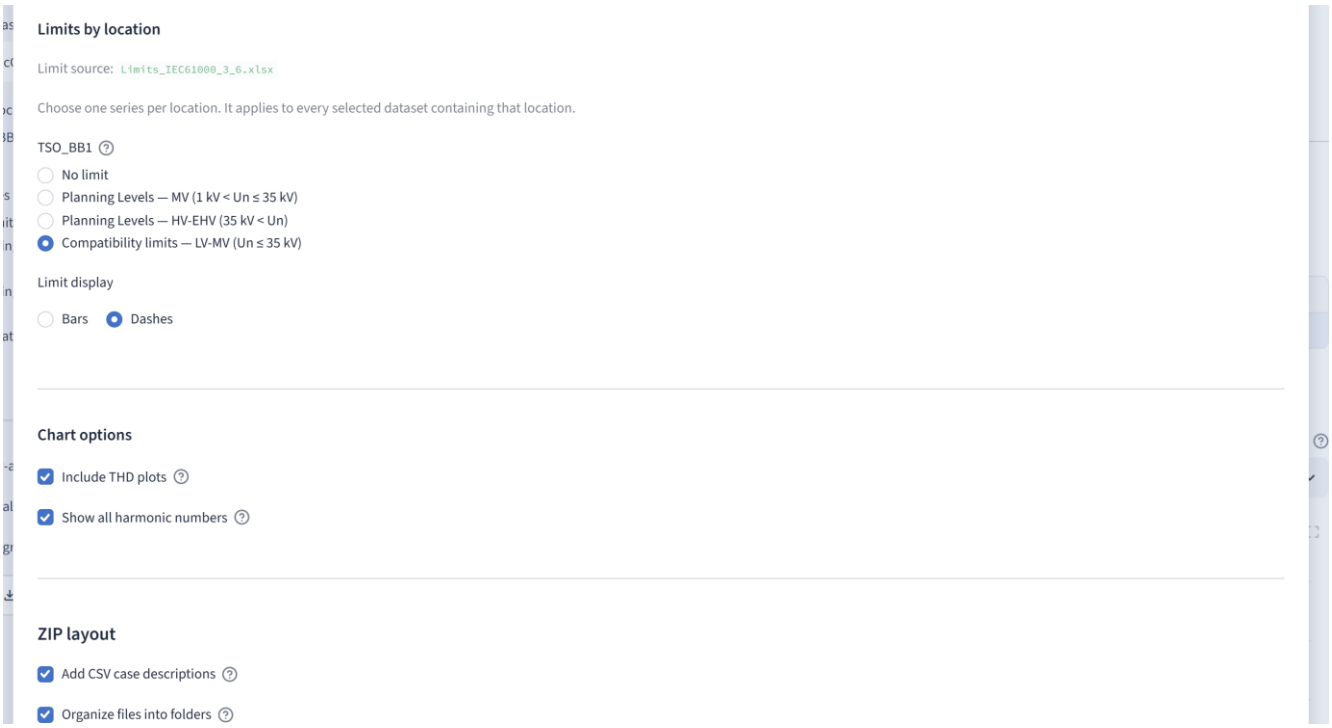
Only rules used for exported custom charts are included in the ZIP root as **custom_selections.json**. The rule can then be imported into another session. A case-description CSV in a custom chart set lists the selected matching cases.

12.4 Set limits and chart options

Choose one **Limit source** for the whole export. Then choose one **Limit series** for each location. That location's series applies to every selected dataset that contains the location. Choose **No limit** where appropriate. **Limit display** chooses **Bars** or **Dashes** for the exported charts.

Include THD plots and **Show all harmonic numbers** are selected by default. These options apply to exported charts. They are independent of the corresponding live chart display.

Figure 10 — Limit series by location, chart options and ZIP layout.



12.5 Set export y-axis limits

Export **Auto y-axis limits** is one mode switch for the complete ZIP and is on by default. Automatic maxima are calculated separately for each dataset. Charts from the same dataset use consistent maxima. Different datasets do not have to share one scale.

Turn Auto off to set a **Spectrum maximum (%)** and a **THD maximum (%)** for each selected dataset. The THD maximum appears when THD charts are included. These values apply across the dataset's selected locations, groups and custom charts. Export maxima do not change the live charts. If a maximum clips data, the dialog reports it; the source data remains unchanged.

12.6 Choose ZIP contents and folders

Add CSV case descriptions is selected by default. With mapping, the ZIP contains a **case_descriptions.csv** for each chart set, with **Case** and **Description** columns. Without mapping, it uses **Add CSV case list** and leaves descriptions blank.

Organize files into folders is selected by default. It creates folders in this order:

dataset / location / group /

When you clear this option, filenames identify the dataset, location, group and chart type. The dashboard normalises names and adds a suffix if output paths collide.

The ZIP can contain:

- harmonic_spectrum.png for each chart set.
- voltage_thd_orders_2-50.png where THD is enabled and complete data allow the chart.
- case_descriptions.csv when case CSV output is enabled.
- custom_selections.json when saved rules produce exported charts.

The ZIP does not contain the plotted harmonic-data CSV.

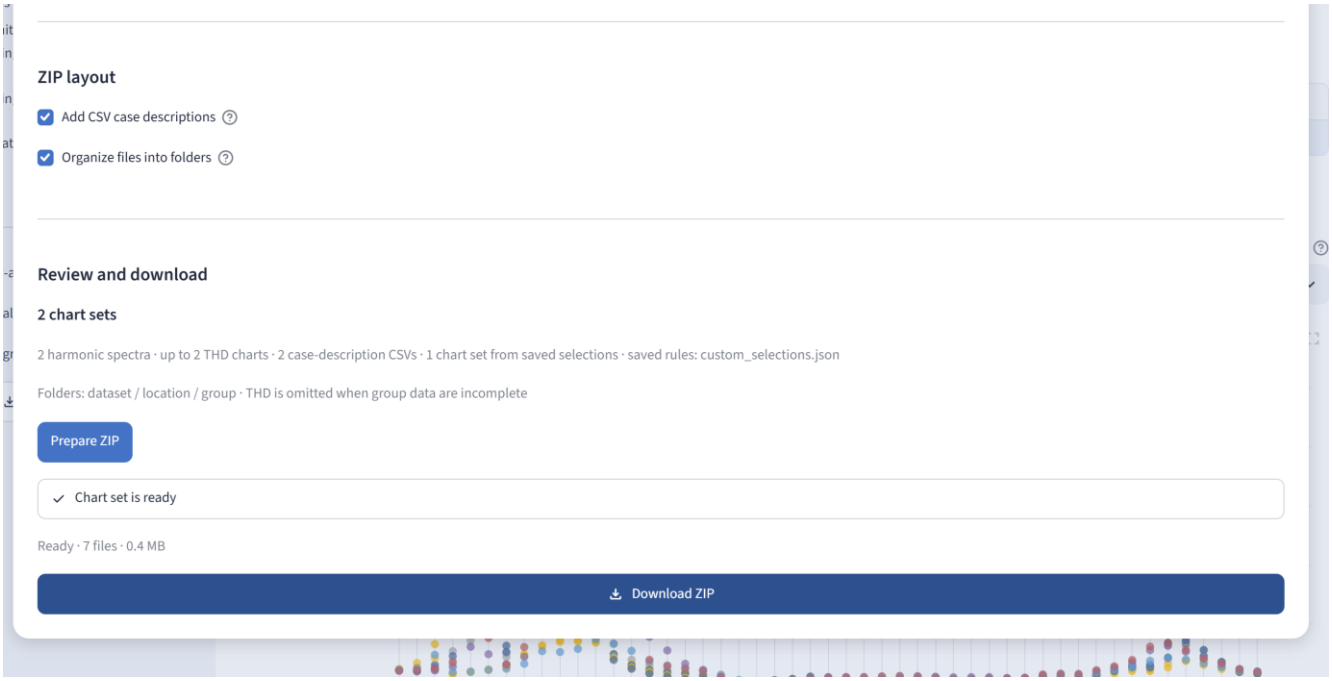
12.7 Review, prepare and download

Read Review and download before preparing the archive. It reports the number of chart sets, harmonic spectra, maximum possible THD charts, case CSVs, custom sets and folder layout. THD images can be omitted when the data are incomplete.

1. Select Prepare ZIP.
2. Follow the progress and status message.
3. When the dialog reports that the chart set is ready, select Download ZIP.

Changing the plan invalidates a prepared archive. Prepare it again after any change. An input error in a selected dataset or location can block the ZIP; correct the workbook or remove the invalid selection. The prepared file remains in the current session. Download it before the session ends.

Figure 11 — Review, preparation status and the ready-to-download archive.



12.8 Image dimensions and hosted rendering

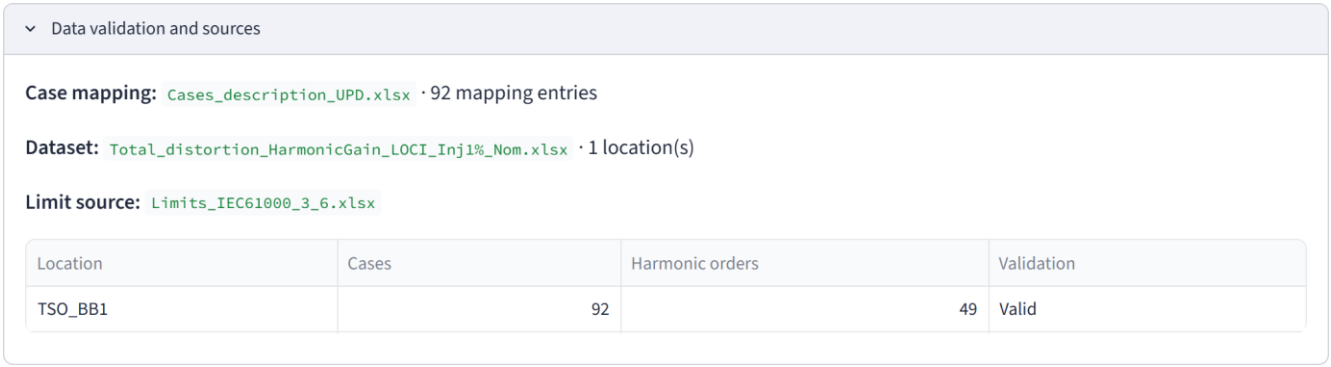
Batch chart PNGs use a fixed 1400 × 505 CSS-pixel design at 2× scale. Each image is 2800 × 1010 pixels. The output dimensions do not change with the user's screen resolution.

The chart-rendering service runs where the app executes. A hosted user does not need to install a separate local rendering tool. If preparation reports that chart rendering is unavailable, keep the message and report it to the application owner. Do not treat this message as a workbook-format error without checking the details.

13. Validation and Troubleshooting

Open **Data validation and sources** to see the active mapping, dataset and limit sources. The location table reports case counts, harmonic-order counts and validation status. Review any warnings or errors before changing workbooks.

Figure 12 — Validation summary for the current result location.



Symptom	Why it happens	What to do
No chart appears	No valid result workbook is loaded.	Upload a valid result .xlsx workbook or a ZIP that contains one.
Mapping workbook error	The required worksheets or columns are missing, or case numbers are duplicated.	Check Cases and Conditions, use the exact required column names, and keep one active mapping workbook.
Cases appear under Unmapped	Result case codes are missing from the mapping workbook.	Add exact matching case numbers when you need descriptions or groups. Otherwise, analyse them under Unmapped or All.
A location cannot be plotted	Its result worksheet has invalid harmonic orders, duplicate headers, or another structural error.	Correct that worksheet. Other valid locations can remain usable.
No limit appears	No valid limits source or series is selected.	Choose No limit, or correct and upload a valid limits workbook. A missing limit is not zero.
A custom rule returns no cases	Its phrases do not match descriptions at the current dataset and location.	Check the description text and ALL, ANY and NOT conditions. Clear or edit the rule when needed.
A case has no THD bar	At least one required order or numeric value is missing.	Review the missing-order message and source workbook. A missing bar does not mean THD equals zero.
A chart value is clipped	A fixed y-axis maximum is below one or more values.	Enable Auto y-axis limits or increase the relevant maximum.
ZIP preparation fails	A selected input is invalid, no valid chart set remains, or rendering is unavailable.	Read the message in the dialog. Correct the input or remove it. Report rendering errors to the application owner.
Prepared ZIP is unavailable	The plan changed or the session ended.	Select Prepare ZIP again and download the archive in the current session.
Saved rules are missing in a new session	Saved rules are session-only.	Import the JSON backup with Upload → Import selections.
A filename conflict appears	Two uploaded workbooks have the same basename, including files inside ZIP folders.	Rename or remove one workbook, then upload the files again.
ZIP upload is rejected	The archive is corrupt, encrypted, or above an archive limit.	Rebuild the ZIP and keep it within the entry-count and uncompressed-size limits in Section 4.5.

An error blocks the affected operation. A warning identifies a condition that may still allow the spectrum to be viewed. For example, unmapped cases remain numerical results, but they have no mapping description. A structural error can block one result location while other valid locations remain available. Invalid or missing limits can be bypassed by choosing **No limit**.

An unavailable or sleeping hosted app is a hosting or connection condition. It is separate from workbook validation.

14. Reference and Limitations

14.1 Control reference

Control	Purpose and scope	Starting state
Add Excel workbooks	Upload .xlsx files or ZIP bundles for this session.	Empty when no uploads exist.
Clear uploaded files	Remove session uploads; keep preloaded project files.	Shown when uploads exist.
Reload project workbooks	Re-scan preloaded inputs and clear related cache; keep session uploads.	Available in the sidebar.
Result dataset	Choose one result workbook.	First valid available dataset.
Busbar / location	Choose one worksheet in that dataset.	First valid available location.
Limit source	Choose a limits workbook when more than one source is available.	Current or first available source.
Limit series	Choose one named curve or No limit.	No limit, unless a session choice is remembered.
Limit display	Show a selected curve as Bars or Dashes.	Bars when a limit is active.
Auto y-axis limits	Let charts fit values or show manual maxima.	On.
Spectrum maximum (%)	Set the live spectrum y-axis maximum.	Hidden while Auto is on.
THD maximum (%)	Set the live THD y-axis maximum.	Hidden while Auto is on.
Show all harmonic numbers	Label every supplied harmonic order.	On.
Show groups as tabs	Show one group view at a time or stack group sections.	On.
Cases to compare	Select plotted cases.	All eligible cases selected.
Select all / Clear all	Select or clear the cases in a multiselect.	Use when required.
Selected case descriptions	Review the current mapped case codes and descriptions.	Available when descriptions exist.
Download plotted data (CSV)	Download the current selected harmonic data.	Available below the view.
New selection	Open a custom-rule editor.	Available on Custom selections.
Save selection / Cancel	Save the current rule or discard the unsaved edit.	In the custom-rule editor.
Edit	Open the selected saved rule.	Requires a selected row.
Restore matches	Restore all cases that match the active rule.	Requires an active rule.
Delete	Delete the selected rule after confirmation.	Requires a selected row.
Clear active	Clear the active rule without deleting it.	Requires an active rule.

Control	Purpose and scope	Starting state
Download	Save the current rule library as JSON.	Available in Saved selections.
Upload / Import selections	Import a valid JSON rule library, replacing the current session library.	Available in Saved selections.
Download charts	Open the batch chart-export dialog.	Available in the sidebar.
Prepare ZIP	Build the archive from the reviewed export plan.	Select after reviewing the plan.
Download ZIP	Save the prepared archive.	Available when preparation succeeds.
Data validation and sources	Review source files and validation by location.	Collapsed or expanded by the user.

14.2 Definitions

Term	Meaning in this guide
Dataset	One result workbook.
Busbar / location	One result worksheet.
Case	A result column identified by its case code.
Case group	The value in the mapping workbook's Group column.
Spectrum	Harmonic voltage distortion plotted separately for each harmonic order.
THD	Root-sum-square of the defined harmonic-voltage percentages from order 2 through H, where H is no higher than 50.
Limit series	A user-selected curve of limit values aligned by harmonic order.
Saved selection	A reusable Boolean rule that searches case descriptions. It is not a static case-code list.
Chart set	One exported spectrum and its optional THD chart and case metadata.

14.3 Limitations

- The dashboard does not generate simulations or validate model assumptions.
- It does not select a limit series automatically by voltage level or location.
- It does not replace missing harmonic or limit values with zero.
- It does not interpolate or extrapolate limit values.
- It does not issue a compliance certificate or an overall engineering pass/fail conclusion.
- THD uses the defined range through order 50; the spectrum can include higher orders.
- Saved selections and uploaded workbooks are session-only unless the user retains and reuses the applicable files or JSON backup.
- Case colours identify traces for visual comparison. They do not classify engineering outcomes.
- A fixed y-axis can clip chart values. It does not change calculations or source data.

14.4 Keep this guide current

This guide describes the application's working behaviour and controls. Review it when the app changes. Use the hosted URL on the cover as the user entry point. Do not use a development folder or an old example-workbook location to access the dashboard.